AR 2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Alphabet

NASDAQ : GOOGL · Mature-Company DCF

$4046.57B mkt cap · 12.09B sh · $126.8B cash, $48.5B debt · $126.8B cash vs $48.5B debt (~$78.3B net cash); ~12.09B shares (Class A+B+C), buyback + a small dividend; Revenue $402.8B FY25 (+15%); GAAP op margin 32.0%; FCF $73.3B; capex ~$91B and guided to ~$190B in FY26; Up ~7% YTD and ~38% TTM to ~$335 / ~$4.05T cap; ATH close $402.38 (May-2026), re-rated on owning the full AI stack

22 October 2026 · the qualitative

$334.76

THE CENTRAL QUESTION

Does Alphabet's full AI stack (TPUs→Gemini→Cloud→apps) defend and monetize Search faster than generative-AI assistants erode its query and ad economics?

Alphabet trades at ~$4.05T (~$335, +7% YTD and ~38% TTM) - roughly 54x EV/FCF on $402.8B revenue growing ~15%, 32% operating margins, net cash, $73.3B FCF before a capex ramp to ~$190B. Cloud just re-accelerated to +63% with a ~$462B backlog and the AI stack re-rated the name, but Search is still ~60%+ of revenue and the price embeds the win. The DCF asks whether Search holds its economics through the generative-AI shift while Cloud compounds. The finding lands -15.6% (weighted ~$283): the modal base sits ~14% below spot, so the run priced in the franchise and then some.

PROBABILITY-WEIGHTED EXPECTED VALUE

most likely case (base) -14.5% · 72% at or below spot

$282.54 expected fair value today

-15.6% vs spot $334.76

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
$414.92	$609.49	$895.58	$1,316.37
1.24× spot	1.82× spot	2.68× spot	3.93× spot

WEIGHTING RATIONALE

Ultra Bear 13% AI assistants erode Search economics; ~$104 (-69%).
Bear 25% Search slows; capex caps returns; ~$171 (-49%).
Base 34% Search defends; Cloud compounds; ~$286 (-14%).
Bull 20% AI stack compounds; Cloud re-rates; ~$423 (+26%).
Ultra Bull 8% Full-stack AI platform re-rates; ~$553 (+65%).

Bull (20%) + Ultra Bull (8%) together contribute $128.85 — 46% of the $282.54 expected value despite 28% combined probability. Today's spot ($334.76) sits 18% above the weighted expected; the forward-weighted value reaches spot in about 2 years.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR $103.95

-68.9% vs spot

CAGR +4.8% WACC 9.0%
CAGR +4.8% Prob 13%

AI assistants gut Search economics.

BEAR $171.32

-48.8% vs spot

CAGR +9.6% WACC 8.5%
CAGR +9.6% Prob 25%

Search slows; capex caps returns.

BASE $286.32

-14.5% vs spot

CAGR +13.4% WACC 8.0%
CAGR +13.4% Prob 34%

Search defends; Cloud compounds.

BULL $422.99

+26.4% vs spot

CAGR +16.2% WACC 7.8%
CAGR +16.2% Prob 20%

AI stack compounds; Cloud re-rates.

ULTRA BULL $553.18

+65.2% vs spot

CAGR +18.7% WACC 7.5%
CAGR +18.7% Prob 8%

Full-stack AI platform; multiple re-rates.

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AR 2EB | ALAMEDA RESEARCH 2: ELECTRIC BOOGALOO — LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

Alphabet scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 13% Bear 25% Base 34% Bull 20% Ultra Bull 8% · Weighted expected $282.54 (-15.6% vs spot)

ULTRA BEAR $103.95

-68.9% vs spot

Probability 13%

AI assistants gut Search economics.

WHY 13%

The genuine bear: Search is ~60%+ of revenue, the capex commitment is now ~$190B and 'significantly higher' in 2027, and chatbots are a live substitution threat. 13% weight on the AI shift impairing the core.

WHAT HAPPENS

Generative-AI assistants take real query share and lower ad load: Search growth stalls toward low-single-digit, AI Overviews monetize below legacy CPCs, and the ~$190B capex ramp lands while revenue decelerates - so FCF margin sinks to ~15-17% off the ~18% base. Cloud keeps growing but can't offset a structurally impaired core.

A franchise whose primary cash engine is being disrupted earns a low-growth, ~16x terminal multiple → DCF ~$104 (-69%). The net cash sets a floor but can't offset a Search that loses its economics while capex stays committed.

BEAR $171.32

-48.8% vs spot

Probability 25%

Search slows; capex caps returns.

WHY 25%

Search holds share but the shift compresses ad economics while ~$190B capex weighs on returns - the plausible 'defends but de-rates' path. 25% given the antitrust overhang and the capex trajectory.

WHAT HAPPENS

Search defends volume but ad load and CPCs drift lower as queries route through AI surfaces, growth decelerates to high-single-digit, and the capex ramp holds FCF margin near 18-20% even as Cloud scales. The AI stack works but the blended economics are lower than the past decade's.

DCF ~$171 (-49%) - a still-dominant platform that grows slower and converts less of revenue to free cash flow at ~19x FCF is worth well below today's price; the post-run multiple already prices the win.

BASE $286.32

-14.5% vs spot

Probability 34%

Search defends; Cloud compounds.

WHY 34%

Requires Search to monetize AI at near-parity and Cloud to keep compounding while capex stays productive - consistent with Q1'26 (Search +19%, Cloud +63%, margin doubling). 34% as the central case, below spot.

WHAT HAPPENS

The modal path: Search defends and monetizes the AI transition at roughly parity (Q1'26's +19% argues defense is winning), Cloud keeps compounding off the ~$462B backlog with expanding margins, and FCF margin recovers from the capex trough toward ~24% as the ~$190B spend builds capacity that fills. Revenue compounds low-double-digit to ~$680B.

DCF ~$286 (-14%) - the modal case lands below spot. A franchise growing low-double-digit at a recovering ~24% FCF margin and a ~23x multiple is worth less than today's price; the 2026 run priced the franchise in, leaving no margin of safety at the modal outcome.

BULL $422.99

+26.4% vs spot

Probability 20%

AI stack compounds; Cloud re-rates.

WHY 20%

Cloud's +63% and doubling margin plus a defended, AI-monetized Search make a genuine compounding case credible. 20% weight on the full AI stack delivering on both legs.

WHAT HAPPENS

The full stack pays off: TPUs lower the cost of serving Gemini, Cloud sustains hyper-growth and operating leverage off the backlog, AI Mode/Overviews lift Search monetization above legacy rates, and FCF margin climbs toward ~27% once the capex wave is absorbed. Revenue compounds mid-teens to ~$730B.

DCF ~$423 (+26%) - if Search not only defends but Cloud + the AI stack compound into a second profit engine, the franchise is worth meaningfully above spot at a ~27x multiple; this is the bull the re-rating is paying for.

ULTRA BULL $553.18

+65.2% vs spot

Probability 8%

Full-stack AI platform; multiple re-rates.

WHY 8%

Everything works - Search monetizes AI above parity AND Cloud compounds AND TPUs/Gemini/Waymo add value AND the multiple holds. ~8%, the long-duration tail.

WHAT HAPPENS

Everything compounds: Gemini + AI Mode expand Search economics, Cloud becomes a durable second franchise at scale, Waymo and the model/TPU stack add optionality, and FCF margin reaches ~29% as the capex cycle turns into high-return capacity. Revenue compounds high-teens to ~$790B.

DCF ~$553 (+65%) - Alphabet re-rates as the broadest full-stack AI platform; ~8% probability, the asymmetric upside that owning chips through apps could deliver.

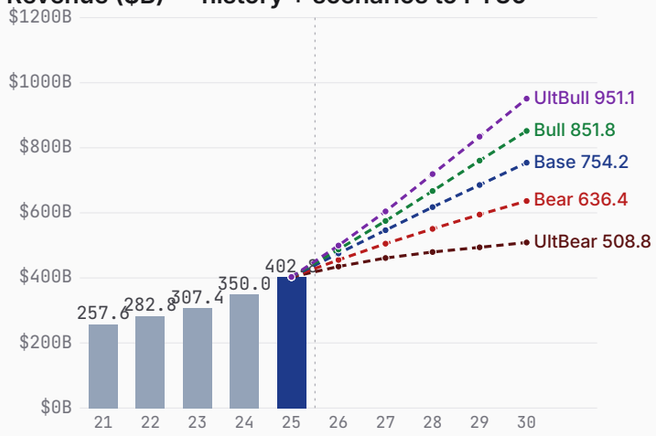
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Alphabet business snapshot

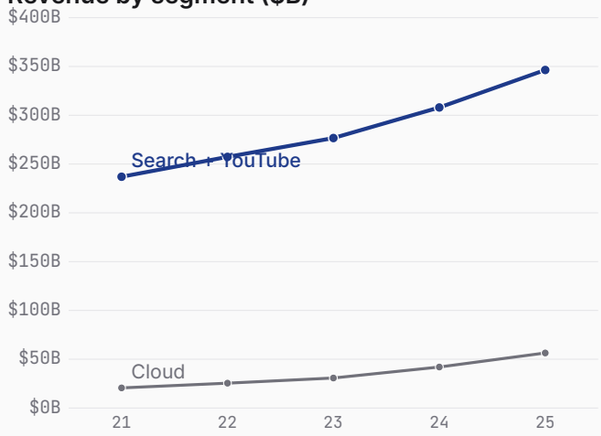
Bear $171.32 Base $286.32 Bull $422.99

FY21-FY25 history + FY26-FY30 scenario projections · fiscal years end Dec · Q1 2026 results

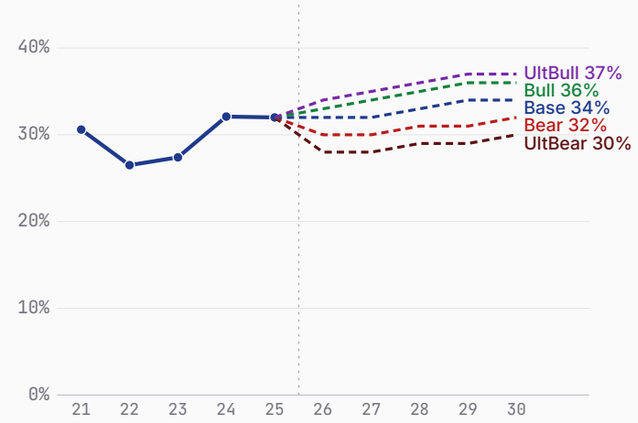
Revenue ($B) — history + scenarios to FY30



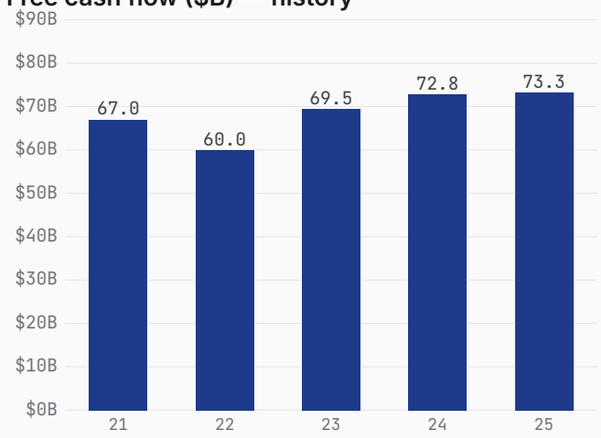
Revenue by segment ($B)



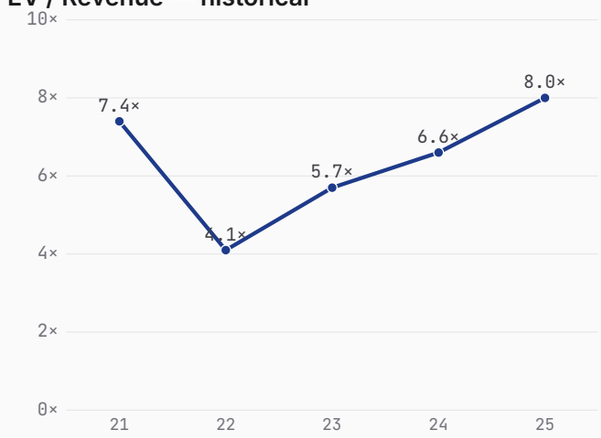
Op margin — history + scenarios



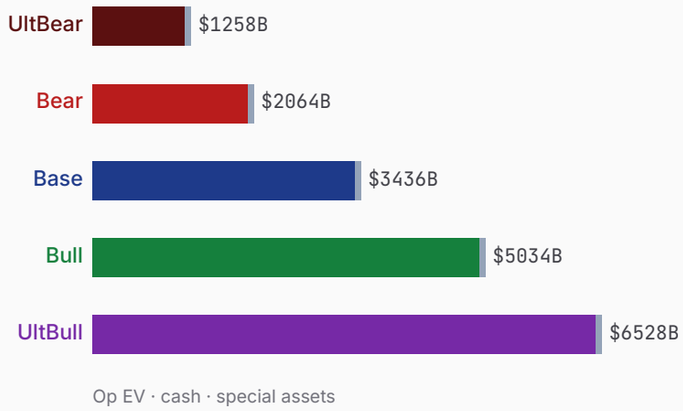
Free cash flow ($B) — history



EV / Revenue — historical



Equity build (Op EV + net cash)



Op EV · cash · special assets

Sources: SEC XBRL company facts (CIK 1652044; revenue, GAAP operating income/margin, cash flow, debt), Alphabet Q1 2026 release (revenue $109.9B/+22%, op margin 36.1%, Search & other +19% to $60.4B, Cloud +63% to $20.0B at 32.9% margin, backlog ~$462B, FY26 capex guided ~$190B). Revenue modeled as a growth-rate path off FY25; FCF = revenue x FCF margin; Gordon terminal at scenario WACC. EV/sales & EV/FCF vs mega-cap platform peers (MSFT/META/AMZN).

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Alphabet 7 Powers · the moat, scored

Bear $171.32 Base $286.32 Bull $422.99

Arena. Search + digital advertising and cloud/AI infrastructure - Alphabet (Search, YouTube, Cloud, the TPU→Gemini→apps stack) vs Microsoft/OpenAI, Amazon, Meta, and chatbot challengers; auditing entrenched network/scale/cornered-resource Powers against a generative-AI substitution and a live antitrust threat.

POWER AUDIT
dominant-power durability: high

POWER AUDIT · 7 POWERS

Scale Economies ■■■	Massive fixed-cost leverage across the index, ad systems, YouTube, and global data centers; 32% operating margin at $403B revenue - few can match the spend per query.
Network Economies · thesis leans here ■■■	Search's two-sided advertiser/user loop, YouTube's creator/viewer flywheel, and Android/Play ecosystems - more users sharpen results and pricing, a genuine network Power.
Counter-Positioning ■□□	The incumbent now being counter-positioned by AI-native assistants (ChatGPT), not the disruptor; AI Overviews is a defensive retrofit, not counter-positioning.
Switching Costs ■■■	Workspace/Cloud contracts and the ~$462B backlog lock enterprises in; consumer Search switching is low, but Android/Chrome/account defaults raise friction.
Branding ■■■	'Google' is a verb and a default trust signal, but brand alone won't hold queries if an assistant answers better - durable yet not the binding Power.
Cornered Resource ■■■	The crawl index, decades of query/click data, custom TPUs, and frontier Gemini models - assets rivals cannot quickly replicate; the data + silicon edge is the deepest moat.
Process Power ■■□	Deep ML/infra engineering capability, but increasingly a contested field rather than a unique repeatable process.

PEER SET

Microsoft / OpenAI — 16% gr · 45% mgn · ~35x P/E
Azure + Copilot + OpenAI partnership attack both Cloud and Search via AI assistants; the primary full-stack rival.

Amazon (AWS) — 15% gr · 11% mgn · ~30x EV/FCF
Cloud market leader; the EV/FCF and Cloud-growth benchmark Alphabet's Cloud is measured against.

Meta Platforms — 20% gr · 41% mgn · ~24x EV/FCF
Ad-budget rival and AI-compute peer; competes for advertiser dollars and AI talent/capex.

OpenAI / Anthropic — · private
Frontier-model labs whose assistants are the direct generative-AI substitution threat to Search query volume.

THREAT VECTORS · FALSIFIERS

Network Economies · ChatGPT / AI-native assistants
falsifier: Search & other growth decelerates toward low-single-digit or query volume visibly leaks to chatbots, breaking the user/advertiser loop.

Cornered Resource · Frontier labs + merchant silicon (NVIDIA)
falsifier: Rivals match Gemini quality and serving cost, eroding the data/TPU advantage as a differentiator.

Scale Economies · DOJ / ad-tech antitrust remedy
falsifier: An adverse ad-tech ruling forces an AdX/DFP divestiture or default-distribution bans dent Search's scale and reach.

COMPETITIVE READ

Alphabet's network and scale Powers are deeply entrenched - a two-sided Search/advertiser loop, YouTube's flywheel, and a cornered-resource stack (index, query data, custom TPUs, Gemini) few rivals can replicate. The Audit: network economies are durable (high); the live threats are generative-AI assistants substituting for ~60%+-of-revenue Search and an antitrust overhang (DOJ search appeal, a pending ad-tech divestiture) the DCF doesn't price. Q1'26 (Search +19%, Cloud +63%) says the defense is winning - but the ~38% TTM run re-rated it to ~54x EV/FCF, so the modal case lands below spot: -15.6%, a durable franchise priced for the win.

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Alphabet show your work

Bear $171.32 Base $286.32 Bull $422.99 · Weighted $282.54 (-15.6%)

ULTRA BEAR $103.95

ASSUMPTIONS

5y revenue CAGR (%)	+4.8
Year-5 op margin (%)	30.0
WACC (%)	9.0
Terminal growth (%)	2.5
Terminal FCF ($B)	$86.49B
Starting revenue ($B)	402.80
Scenario probability (%)	13

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	435.02	+28%	+65.25	+59.87
FY27	461.13	+28%	+69.17	+58.22
FY28	479.57	+29%	+76.73	+59.25
FY29	493.96	+29%	+83.97	+59.49
FY30	508.78	+30%	+86.49	+56.21

Σ PV explicit FCF	**+293.04**
+ PV terminal (g 2.5%)	886.45

EQUITY BUILD · $B & PER SHARE

Operating EV	$1179.49B
+ Cash & investments	$126.80B
− Total debt	$48.50B
= Equity value	**$1257.79B**
FY30 shares (M)	12,100
= Expected per share	**$103.95**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$159.94	$246.09	$378.64	$582.58
0.48×	0.74×	1.13×	1.74×

BEAR $171.32

ASSUMPTIONS

5y revenue CAGR (%)	+9.6
Year-5 op margin (%)	32.0
WACC (%)	8.5
Terminal growth (%)	3.0
Terminal FCF ($B)	$127.28B
Starting revenue ($B)	402.80
Scenario probability (%)	25

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	455.16	+30%	+77.38	+71.32
FY27	505.23	+30%	+90.94	+77.25
FY28	550.70	+31%	+104.63	+81.92
FY29	594.76	+31%	+118.95	+85.83
FY30	636.39	+32%	+127.28	+84.65

Σ PV explicit FCF	**+400.96**
+ PV terminal (g 3.0%)	1585.18

EQUITY BUILD · $B & PER SHARE

Operating EV	$1986.14B
+ Cash & investments	$126.80B
− Total debt	$48.50B
= Equity value	**$2064.44B**
FY30 shares (M)	12,050
= Expected per share	**$171.32**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$257.61	$387.35	$582.44	$875.80
0.77×	1.16×	1.74×	2.62×

BASE $286.32

ASSUMPTIONS

5y revenue CAGR (%)	+13.4
Year-5 op margin (%)	34.0
WACC (%)	8.0
Terminal growth (%)	3.5
Terminal FCF ($B)	$181.00B
Starting revenue ($B)	402.80
Scenario probability (%)	34

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	475.30	+32%	+90.31	+83.62
FY27	546.60	+32%	+109.32	+93.72
FY28	617.66	+33%	+135.88	+107.87
FY29	685.60	+34%	+157.69	+115.91
FY30	754.16	+34%	+181.00	+123.18

Σ PV explicit FCF	**+524.30**
+ PV terminal (g 3.5%)	2833.24

EQUITY BUILD · $B & PER SHARE

Operating EV	$3357.54B
+ Cash & investments	$126.80B
− Total debt	$48.50B
= Equity value	**$3435.84B**
FY30 shares (M)	12,000
= Expected per share	**$286.32**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$420.70	$618.14	$908.26	$1,334.53
1.26×	1.85×	2.71×	3.99×

BULL $422.99

ASSUMPTIONS

5y revenue CAGR (%)	+16.2
Year-5 op margin (%)	36.0
WACC (%)	7.8
Terminal growth (%)	4.0
Terminal FCF ($B)	$229.99B
Starting revenue ($B)	402.80
Scenario probability (%)	20

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	487.39	+33%	+97.48	+90.42
FY27	575.12	+34%	+126.53	+108.88
FY28	667.14	+35%	+160.11	+127.81
FY29	760.54	+36%	+197.74	+146.43
FY30	851.80	+36%	+229.99	+157.98

Σ PV explicit FCF	**+631.52**
+ PV terminal (g 4.0%)	4323.72

EQUITY BUILD · $B & PER SHARE

Operating EV	$4955.24B
+ Cash & investments	$126.80B
− Total debt	$48.50B
= Equity value	**$5033.54B**
FY30 shares (M)	11,900
= Expected per share	**$422.99**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$615.78	$896.43	$1,305.00	$1,899.79
1.84×	2.68×	3.90×	5.68×

ULTRA BULL $553.18

ASSUMPTIONS

5y revenue CAGR (%)	+18.7
Year-5 op margin (%)	37.0
WACC (%)	7.5
Terminal growth (%)	4.0
Terminal FCF ($B)	$275.81B
Starting revenue ($B)	402.80
Scenario probability (%)	8

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	499.47	+34%	+104.89	+97.57
FY27	604.36	+35%	+145.05	+125.51
FY28	719.19	+36%	+186.99	+150.52
FY29	834.26	+37%	+233.59	+174.91
FY30	951.06	+37%	+275.81	+192.12

Σ PV explicit FCF	**+740.63**
+ PV terminal (g 4.0%)	5708.58

EQUITY BUILD · $B & PER SHARE

Operating EV	$6449.21B
+ Cash & investments	$126.80B
− Total debt	$48.50B
= Equity value	**$6527.51B**
FY30 shares (M)	11,800
= Expected per share	**$553.18**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$794.16	$1,140.12	$1,636.79	$2,349.83
2.37×	3.41×	4.89×	7.02×

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Alphabet People · Opportunity · Context · Deal

Bear $171.32 Base $286.32 Bull $422.99

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Sundar Pichai

6 yrs in seat

6%
insider ownership

MEDIUM
key-person risk

Capital allocation (realized)
Capital return ramping off a long no-payout history: initiated the first-ever dividend April 2024 ($0.20/qtr, raised 5% to $0.21 in 2025), alongside two consecutive $70B buyback authorizations (April 2024 + April 2025). Simultaneously a record capex build — FY2025 guidance raised $75B to $85B to $91-93B for AI infrastructure (TPUs/GPUs, data centers). Selective tuck-in M&A; SBC a multi-billion annual cost.

Incentive alignment
Non-founder CEO paid mostly in equity: FY2024 total comp $10.73M on a $2M base, the remainder equity. PSUs granted on a ~3-year cycle (2024 was an off-grant year; the 2022 grant pushed total comp to ~$226M); long-term grants carry performance/TSR conditions. Pichai holds Class A/C from comp, not the super-voting Class B.

GOVERNANCE FLAGS

- triple-class: Class A (1 vote, GOOGL) · Class B (10 votes, founder-held, unlisted) · Class C (0 votes, GOOG)
- founders Page + Brin control ~52.7% of votes via Class B (Page 27.1% + Brin 25.2%) on ~6% economics — majority voting control
- CEO Pichai operates but holds no Class B; founders, not the CEO, hold control and cannot be outvoted by outside holders
- public Class C (GOOG) carries no vote at all

THE FOUR LEGS



People
observable composite · 1–5

Opportunity
The scenario distribution · the business snapshot · the Arthur Indicator

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market common — entry price weighed against the probability-weighted fair value, then position-sized.
-15.6% vs spot $334.76

> **PEOPLE READ**
> Founders Page and Brin retain ~52.7% combined voting control via 10-vote Class B on roughly 6% economic ownership, while non-founder CEO Pichai (Alphabet CEO since Dec 2019) runs operations; capital return is ramping (first dividend 2024, repeated $70B buybacks) against a record AI capex build.

Alphabet supporting analysis · disclaimers · glossary

Bear $171.32 Base $286.32 Bull $422.99

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **The run priced the franchise in.**
After +7% YTD / ~38% TTM the name is ~54x EV/FCF and the modal DCF is ~-14% - the AI stack is real, but the price already embeds the win, leaving no margin of safety.

2 **Cloud at +63% with doubling margins is the new engine.**
Cloud hit $20.0B/+63% at 32.9% operating margin (from 17.8%) with a ~$462B backlog ~2x QoQ - a credible second franchise that the bull/ultra-bull cases lean on.

3 **Search defense is winning so far.**
Q1'26 Search & other still grew +19% to $60.4B with AI Overviews at 2.5B MAU monetizing near parity - the central bear (query/CPC leakage) is not yet visible in the numbers.

4 **The ~$190B capex is the FCF swing.**
Capex guided to ~$190B FY26 and higher in 2027 crushes near-term FCF; the bull needs it to convert to high-return capacity.

5 **Antitrust is the unpriced tail.**
The DOJ search remedy (default bans + data-sharing, on appeal) and a pending ad-tech remedy (possible AdX/DFP divestiture) sit outside the DCF.

FALSIFICATION TRIGGERS

Bull validation

Cloud growth holds >40% with margin above 30% · Search & other stays >+15% as AI Overviews/AI Mode monetize · FCF margin recovers above ~22% as capex is absorbed · backlog keeps building

Bear validation

Search & other decelerates toward low-single-digit or CPCs fall · FCF margin stays below ~18% as ~$190B+ capex outruns cash · Cloud growth/margin rolls over · an adverse ad-tech (AdX/DFP) divestiture remedy

Reframe needed

if generative-AI assistants take durable query share and ad load drops, the ~60%-of-revenue Search engine loses its economics - re-rate the terminal toward a slower-growth, lower-FCF-margin platform

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AI-assisted analysis.

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Search & other — Alphabet's core search-advertising revenue (~$60.4B in Q1'26, ~60%+ of total); the swing-factor segment the generative-AI shift threatens.

TPU — Tensor Processing Unit - Google's custom AI accelerator; serves Gemini and Cloud AI below merchant-silicon cost.

Google Cloud — The cloud-infrastructure + Workspace segment ($20.0B/+63% in Q1'26 at ~33% operating margin, ~$462B backlog) - the second engine.

FCF margin — Free cash flow / revenue (~18% FY25, depressed by the capex ramp); whether ~$190B capex converts to ~24% recovery.

AI Overviews / AI Mode — Gemini-powered answers atop Search (2.5B / 1B MAU) monetizing roughly at Search-parity - how Google defends ad economics through the AI transition.